

Paul Abrams

Toronto, Ontario

Senior Cloud Application Specialist / Engineer · Microsoft Azure (App Service, Functions, AKS) · API-Driven Automation of Manual Processes · End-to-End CI/CD & DevSecOps · Docker / Kubernetes / OpenShift · Azure DevOps · Jenkins · Ansible · Terraform · 15+ Years

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Professional Summary

Senior cloud application specialist with 15+ years designing, building, and operating cloud-native and distributed enterprise systems across the Government of Canada — most recently a **cloud-native microservices** application on **Microsoft Azure (App Service, Azure Functions, AKS)**.

Signature pattern: replacing manual processes with code that drives RESTful APIs — from **Terraform** infrastructure-as-code that provisions cloud infrastructure and IP addresses, to **PowerShell / TypeScript** tools calling REST APIs that deleted over-engineered legacy systems, to a script that **queries the SonarQube REST API to automate vulnerability triage into Azure DevOps**. Designs and consumes **REST APIs between services** and governs them through an **API gateway (Azure API Management)** with authentication, throttling, and transformation. **DevSecOps end-to-end:** designs and operates **CI/CD pipelines with integrated security** — **SonarQube** (SAST) and **OWASP ZAP** (DAST) in-pipeline, **CWE / CVSS-graded** audits, PR-gated release gates, and **ITSG-33 / PBMM** controls — with a track record of migrating downstream identity to **Azure Managed Identities** to eliminate a whole class of secret-rotation risk. Hands-on across **Docker, Kubernetes/AKS, Red Hat OpenShift, Jenkins, and Ansible**, with **enterprise cloud architecture, data engineering (ETL/ELT, Azure Data Factory), and observability (Azure Monitor, Application Insights, Log Analytics)**. Deep back-end in **.NET Core / C#, Node.js / NestJS**, with **Java, Python, SQL, and TypeScript**. **Professional Scrum Master I**. Active open-source contributor — **60+ public PRs** including Microsoft's `vsts-team-calendar`, `tfs-cli`, and the **Microsoft Authentication Library .NET docs**.

Professional Experience

Abrams Software Toronto, Ontario (Remote)

August 2025 – Present

Cloud Application Developer · Independent

- **TWNR** — **architected and shipped a public-facing multiplayer game as a full-stack web application, solo**. Dockerized **TypeScript** monorepo: **Node.js (Express) + WebSocket** backend backed by **PostgreSQL**, per-player **JWT auth**, and a **JSON-Schema-generated REST API contract** that doubles as live documentation. Concurrency-correct state via PostgreSQL transactions (`SELECT ... FOR UPDATE`). Live at twnr.paulabrams.ca.
- **Chess Attack Visualization** — **React 19 + TypeScript** app consuming the **Lichess.org REST API**, with a **NestJS** (Spring Boot-style decorator framework) auth proxy implementing the **OAuth2 PKCE** flow. Diagnosed a missing feature in `chess.js`, opened [upstream PR #572](#), and published a fork on npm. **Portfolio site** operated as **Astro on GitHub Pages with GitHub Actions CI/CD**.
- **LLM code evaluation** for [DataAnnotation.tech](#) and `Stellar.ai` — prompt fixtures and grading rubrics across **.NET / C#, Java, Python, TypeScript, React, Node / NestJS, Go, Rust, SQL, and MongoDB**, including **security audits with CWE / CVSS-graded inventories**. **60+ public PRs** to outside repos including the **Microsoft Authentication Library .NET docs** and `canada-ca/platform-security`.

Employment and Social Development, Canada Gatineau, QC

December 2021 – July 2025

Senior Cloud Application Engineer — Cloud-Native Microservices (Benefits Delivery Modernization)

- Enhanced and maintained a **cloud-native microservices** enterprise onboarding application — **.NET Core / C# back end, Blazor front end, Azure SQL, Azure Functions, GraphQL** — serving procurement, HR, and IT-ops personnel. **Designed and consumed REST APIs between services** and

published and governed them through **Azure API Management (API gateway)** with authentication, throttling, and transformation policies; integrated enterprise IAM via **Azure AD / Entra (OAuth 2.0, OpenID Connect)**.

- **Replaced manual processes with API-driven code:** wrote **Terraform** (infrastructure-as-code) to provision cloud infrastructure — **including IP-address allocation** — through provider REST APIs instead of manual configuration. Built **durable Azure Functions** that called the **Azure DevOps REST API** to create work items, run pipelines, and provision users in on-prem Active Directory (via AAD Connect) by personnel profile — **bulk submissions fanning out asynchronously across the microservices**, replacing manual onboarding steps.
- **End-to-end CI/CD with integrated security (DevSecOps):** operated **Azure DevOps pipelines** for build, **Docker** containerization, **Kubernetes / AKS** deployment, and **Azure Data Factory** (ETL/ELT) orchestration, with **SonarQube** (SAST) quality/security gates and PR-gated release controls. Completed formal **Red Hat OpenShift** training and **resolved OpenShift production-support tickets**; troubleshooted an **Ansible**-driven provisioning pipeline for BDM onboarding-training environments; evaluated **Aqua** container security. Ensured pipelines met **ITSG-33 (NIST-based) controls** against a **PBMM** profile.
- **Enterprise security integration:** migrated the application's downstream identity from rotating secrets (Key Vault + ADO PATs on 90-day rotation) to **Azure Managed Identities** — eliminating the entire secret-rotation incident class and properly attributing automated activity. Managed **TLS / certificate hygiene** and **Key Vault** secrets.
- **Observability:** instrumented the Azure workloads with **Application Insights, Azure Monitor, and Log Analytics**; connected **Power BI to ADO Analytics** for delivery metrics. Shipped **custom web extensions in React / TypeScript / Node.js** against the ADO Builds REST API and mentored junior developers in weekly demos and design walkthroughs.

Employment and Social Development, Canada Gatineau, QC August 2019 – November 2021

Development Lead, Testing Services — CI/CD Automation & DevSecOps

- **Signature pattern — code against REST APIs to replace manual work:** co-designed a **REST API** bridging an internal **C# / .NET** scenario-authoring web app with **Azure DevOps pipelines** (versioned script-fetch and structured results-push endpoints), and wrote a script that **queried the SonarQube REST API to flag vulnerabilities by keyword and surface them into Azure DevOps** — automating a manual security-review step. Worked alongside a **Jenkins**-based automated-testing helper app under development by the team.
- **DevSecOps in the pipeline:** integrated **SonarQube** static analysis (SAST) and **OWASP ZAP** dynamic scans (DAST) into Azure DevOps pipelines; evaluated **MicroFocus Fortify / ScanCentral** (SAST) for adoption. Engineered pipelines that provisioned Windows build agents for **parallelized automated runs** against deployed applications, feeding structured results back into work items.
- Drove a tooling consolidation (**LoadRunner → JMeter**) that **cut software licensing costs** while preserving capability; performed **code review and standards enforcement** and contributed to performance-optimization analysis of high-traffic applications.

Department of Immigration, Refugees, and Citizenship, Canada

Ottawa, ON

August 2017 – August 2019

Team Lead / Technical Lead — Platform & Tooling

- **Technical leadership of a cross-functional team** — led **Agile / Scrum** sprint planning, review, retrospective, daily stand-up, and backlog refinement; introduced poker-planning estimation; set engineering standards and drove delivery cadence.
- Implemented and deployed **custom web extensions in React / TypeScript / Node.js**; mentored **developers in SQL Server query optimization** — query tuning, indexing strategy, partitioned tables, and execution-plan analysis. Administered departmental **Azure DevOps Server / TFS** instances, customized process templates, and managed source-control security across multiple project collections.

Department of Justice, Canada Ottawa, ON

December 2015 – August 2017

Acting Lead Developer — iCase / Microsoft Dynamics CRM Modernization

- **Lead developer** on the live **Legal Case Management System (iCase)** — a large **n-tier ASP.NET / C# / ADO.NET** enterprise web application used by **5,000+ lawyers** — while concurrently building its **Microsoft Dynamics CRM** successor. Modified schema and **front-end UI to surface data-migration status in real time**, and adjusted integration stored procedures consumed by the department's SAP-based financial system.
- Engineered **Dynamics CRM plug-ins, custom actions, and workflows in C#** (rewriting legacy stored procedures as C# / LINQ); used **SSIS** for **ETL** between iCase and Dynamics; produced **entity-relationship diagrams** and **effort-estimation breakdowns (FDD / TDD)**. **Established code-review processes gated on work items** and **mentored junior developers**.

Department of Citizenship and Immigration, Canada Ottawa, ON April 2014 - December 2015

Senior Software Engineer · Release Engineering Lead

- **Replacing manual processes via code and APIs:** hired alongside one other engineer to take over a release-management environment from a contractor team. **Decommissioned over-engineered legacy C# desktop apps** and replaced them with lightweight **PowerShell calling REST APIs** — most contractor tooling was simply deleted and **everything ended up in source control**.
- Implemented and deployed **web extensions in React / TypeScript / Node.js**; maintained a fork of Microsoft's `vsts-team-calendar` extension and contributed back upstream. **Installed, configured, customized, and upgraded** the Azure DevOps Server / TFS platform across dev / UAT / production — complex multi-step upgrades, **Kerberos / SPN configuration**, and database-tier migration.

Department of Justice, Canada Ottawa, ON October 2008 - April 2014

Full-Stack Developer — Legal Case Management System (iCase)

Core developer on the department's flagship **Legal Case Management System (iCase)** — a **distributed, multi-tier ASP.NET / C# enterprise web platform** used by **5,000+ government lawyers and contracted agent firms across Canada**, deployed across **5 load-balanced web servers** and **11 regionally distributed document servers** with **24/7 high availability**.

- **Optimized a high-traffic backend:** diagnosed a homepage report taking 10 s uncached — profiled with SQL Profiler, Report Execution logs, and execution plans, then replaced live joins with a nightly-populated flat table for a **50x speedup uncached (3-4x cached)**. Built a **self-initiated availability-monitoring suite** across the distributed web and document servers, and diagnosed and resolved a **COM Interop memory leak** on the document tier.
- **Java automation-framework development:** architected a **Java-based framework (IBM Rational Functional Tester)** that **parsed English-language scenario documents into executable scripts with round-trip code generation**, plus a companion manager in **C# / SQL Server** — replacing a **two-week, 20-25-person manual pre-release validation cycle** with automated execution reviewed by one person in five days (**98% reduction in effort, \$160K saved per release cycle**).

Education

Bachelor of Computer Science (Honours) - Carleton University September 1999 - May 2003

Certifications & Awards

Professional Scrum Master I (Scrum.org) · ITIL v3 Foundation (Loyalist Certification Services) · DOJ Team Merit Award 2012; Team Merit Award 2011; Team Spirit Award 2009

Technical Skills

Cloud: Microsoft Azure — App Service, Azure Functions (serverless), AKS, Azure SQL, API Management, Key Vault, Managed Identities, Entra / AAD, Storage, Data Factory · AWS — self-hosted pipeline-agent deployment + foundational training · GCP — TWRN production hosting

API-Driven Automation: writing code to drive RESTful APIs and replace manual processes — **Terraform** IaC (infrastructure & IP provisioning via provider APIs), **Azure DevOps REST API** tooling, **SonarQube REST API** vulnerability automation, **PowerShell / TypeScript** REST clients · REST API design between services, GraphQL, WebSocket · API gateway / management (Azure API Management)

CI/CD & DevOps: end-to-end CI/CD pipeline design · **Azure DevOps Pipelines (YAML)**, **GitHub Actions**, **Jenkins** · **Ansible** (provisioning-pipeline troubleshooting + config management) · **Docker**, **Kubernetes / AKS**, **Red Hat OpenShift** (training + production troubleshooting) · Azure Data Factory (ETL/ELT)

DevSecOps / Security: integrated security in CI/CD — **SonarQube** (SAST, pipeline + API automation), **OWASP ZAP** (DAST), MicroFocus Fortify / ScanCentral & Aqua (evaluated) · **OAuth 2.0 / OpenID Connect / PKCE**, **Entra / AAD IAM**, **Managed Identities** · TLS / certificate management, **Key Vault** secret hygiene, encryption · **ITSG-33 / PBMM controls**, CWE / CVSS-graded audits, compliance (SA&A)

Observability: Azure Monitor, **Application Insights**, **Log Analytics** · Prometheus / Grafana (familiarity) · self-built distributed availability-monitoring suite · Power BI on ADO Analytics

Enterprise Cloud Architecture: cloud-native microservices, event-driven & async messaging (queue- / storage-triggered), serverless, load balancing & high availability across cloud & on-prem tiers, IAM & encryption, design patterns, entity-relationship modelling

Languages & Back-End: .NET Core / C#, Java, Python, SQL (T-SQL), TypeScript, JavaScript, PowerShell, XML / XSLT, Go (light) · **ASP.NET Core**, **ADO.NET**, **Entity Framework**, **LINQ**, **Node.js**, **Express**, **NestJS**, **Spring Boot** (familiarity) · **React 19**, **Blazor**

Databases / Data Engineering: MS SQL Server 2005 → 2019 (T-SQL, stored procedures, indexing, partitioning, execution-plan tuning), **Azure SQL**, **PostgreSQL** · **ETL/ELT** (Azure Data Factory, SSIS) · **MongoDB** (working familiarity)

Process / SDLC: Agile / Scrum (PSM I) · **structured methodologies** for design, development, implementation & maintenance · code review, mentoring, technical documentation · ITIL v3 Foundation